Assignment 4

Week 4

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Collaborators. Claude (for LaTeX styling and grammar checking)

Problem 1. Find Eigenvalues using Colab

Find the eigenvectors and eigenvalues of σ_x using colab. You may use this code. But please ask yourself, which basis are we using when we express σ_x in the following form?

```python
sigmax = np.array([[0, 1], [1, 0]])
print(sigmax)

eigenvalues, eigenvectors = np.linalg.eig(sigmax)
print(eigenvalues)
print(eigenvectors)
```

Solution. The solution goes here.

Problem 2. Construct Projection Operator

Now, use hand calculation, derive the projection operators to project to the eigenstates of σ_z.

Solution. The solution goes here.

Problem 3. Use Projection Operator to find Measurement probability

Use the projector equation to find the probability of measuring spin up and down when it is applied to the eigenvectors of σ_x.

Hints: you should get 50% for all of them.

Solution. The solution goes here.

Problem 4. Vector normalization by hand and using python 2

Check **Problem 3.** using co-lab.

Hints:

- Use `np.transpose` for transpose.
- Use `np.dot` for multiplication, including outer product.
- Use `np.linalg.eig` to find eigenvalues.

Solution. The solution goes here.

Problem 5. Orthonormality of Unitary Matrix Row Vectors

Prove that the row vectors of a unitary matrix are orthonormal.

Solution. The solution goes here.

Problem 6. Finding the Determinants of Pauli Matrices by Hand

Find the determinants of σ_x, σ_y, and σ_z. What do you notice?

Solution. The solution goes here.

Problem 7. Finding the Determinants of Pauli Matrices using Colab

Confirm your answers in **Problem 6.** using Colab. This is a sample code.

```
M = np.array([[1, 0], [0, -1]])
print(np.linalg.det(M))
```

Solution. The solution goes here.

Problem 8. Proof of Unitarity

For an orthonormal set of basis, $|v_i\rangle$, we have $\langle v_i|v_j\rangle = \delta_{i,j}$. We transform it using a matrix A, i.e. they become $A\,|v_i\rangle$. Prove that if after the transformation, they are still orthonormal, then A is unitary.

Hint: Express the basis vectors in column form and each of them has only 1 entry. Express A in its matrix form. The fact that after the transformation, they are still orthonormal implies that the columns of A are orthonormal and thus A is unitary.

Solution. The solution goes here.

APPENDIX 1

Assignment 04 | Code Solution

assignment-04-code

August 28, 2026

1 Assignment 04

1.1 1. Find Eigenvalues using Colab

[ ]:
```

```

1.2 4. Vector normalization by hand and using python 2

[ ]:
```

```

1.3 7. Finding the Determinants of Pauli Matrices using Colab

[ ]:
```

```
